

Quartic Stark Phase Quantization as a Weak-Solution Equivalence

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Abstract

Claim boundary. We isolate two PROVED consequences of Roblot's quartic index-solution theorem. In a cyclic-quartic case where the rank-one Stark conjecture is already proved, the phase ratio between a Stark derivative and any Roblot weak solution is a fourth root of unity. In an uncertified case satisfying the same algebraic hypotheses, that phase quantization is equivalent to the full rank-one Stark conjecture; it is not a finite-precision proof mechanism. We derive the root-of-unity factor from the real-place hypothesis and spell out the even-character, conjugate-character, Fourier-inversion, and abelian-extension steps. No census result is claimed here, and this local draft is not authorized for submission or circulation before the preregistered counterexample screen exists as its data section.

1 Setup and claim boundary

Let K/k be cyclic quartic with

$$G = \text{Gal}(K/k) = \langle \gamma \rangle, \quad \tau = \gamma^2, \quad \chi(\gamma) = i.$$

Let $S \supseteq S(K/k)$ contain a distinguished real place v of k which splits completely in K/k , and fix $w \mid v$. We impose Roblot's hypotheses:

- (A1) k is totally real, the places of K above v are real, and the other infinite places of K are complex;
- (A2) the maximal totally real subfield K^+ has $[K : K^+] = 2$;
- (A3) every finite prime in S is ramified or inert in K/K^+ .

These are exactly the assumptions used in [1, pp. 392–393]. We write U_K for the global units and use the right Galois exponent u^g .

Lemma 1 (Real-place root-of-unity rider). *Under (A1), the root-of-unity group of K is*

$$\mu(K) = \{\pm 1\}.$$

Consequently the denominator $e = |\mu(K)|$ in the rank-one Stark formula is 2.

Proof. The place w gives an embedding $K \hookrightarrow \mathbb{R}$. An embedding is injective and sends a root of unity to a real root of unity. The only real roots of unity are 1 and -1 , both of which lie in K . Thus $\mu(K) = \{\pm 1\}$. $\square$

For any character ψ of G and $u \in U_K$, put

$$c_\psi(u) = \frac{1}{2} \sum_{g \in G} \psi(g) \log |u^g|_w. \tag{1}$$

For $a = s\gamma^j \in \{\pm\gamma^j : 0 \leq j < 4\}$, where $s \in \{\pm 1\}$, define signed right exponentiation and its character value by

$$u^a = (u^{\gamma^j})^s, \quad \chi(a) = s i^j.$$

We separately define the left group-ring action

$$a \cdot u := u^{a^{-1}}. \quad (2)$$

This distinction removes an inverse-convention ambiguity in the earlier project note.

Lemma 2 (Pinned covariance). *For every signed a and every character ψ ,*

$$c_\psi(u^a) = \psi(a)^{-1} c_\psi(u), \quad c_\psi(a \cdot u) = \psi(a) c_\psi(u). \quad (3)$$

Proof. For $a = s\gamma^j$, reindexing $t = r + j$ in (1) gives

$$c_\psi(u^{\gamma^j}) = \frac{1}{2} \sum_r \psi(\gamma)^r \log |u^{\gamma^{r+j}}|_w = \psi(\gamma)^{-j} c_\psi(u).$$

The exponent $s = -1$ changes every logarithm's sign. This proves the first identity; applying it to a^{-1} proves the second. $\square$

Roblot's Theorem 6.1 [1, pp. 404–405] supplies a class $\bar{\eta} \in \bar{U}_K^-$ satisfying his index properties (P1) and (P2), unique under the signed action $\{\pm\gamma^j\}$. We fix a representative η for which

$$K(\sqrt{\eta})/k \quad \text{is abelian.} \quad (4)$$

The same theorem proves

$$|L'_{K/k,S}(0, \chi)| = |c_\chi(\eta)|. \quad (5)$$

This is an absolute-value statement, not the oriented Stark identity.

2 Certified cases

Theorem 3 (Certified-case quantization; PROVED). *Assume (A1)–(A3) and assume the rank-one Stark conjecture has been proved for $(K/k, S, w)$. Let ϵ be a Stark unit in the convention*

$$L'_{K/k,S}(0, \psi) = c_\psi(\epsilon) \quad \text{for every character } \psi \text{ of } G. \quad (6)$$

Then there is a signed element $h \in \{\pm\gamma^j\}$ such that $\bar{\eta} = h \cdot \bar{\epsilon}$, and

$$\boxed{\frac{L'_{K/k,S}(0, \chi)}{c_\chi(\eta)} = \chi(h)^{-1} \in \mu_4.} \quad (7)$$

Proof. The index formulae for a Stark unit give (P1) and (P2) [1, Thm. 2.3]. The uniqueness clause in Roblot's Theorem 6.1 therefore gives $\bar{\eta} = h \cdot \bar{\epsilon}$ for a signed h . Lemma 2 and (6) give

$$c_\chi(\eta) = \chi(h) c_\chi(\epsilon) = \chi(h) L'_{K/k,S}(0, \chi).$$

Since the signed quartic character values are precisely μ_4 , rearrangement proves (7). $\square$

Remark 4 (Interpretation). *Theorem 3 is a formal consequence of a proved Stark identity and Roblot's uniqueness theorem. In an already certified case it is a convention check, not new evidence for algebraicity.*

3 Equivalence in uncertified cases

Theorem 5 (Weak-solution/Stark equivalence; PROVED). *Assume (A1)–(A3) and fix all conventions above before either side is evaluated. The following are equivalent:*

1. *the abelian rank-one Stark conjecture holds for $(K/k, S, w)$;*
2. *there is a signed $h \in \{\pm\gamma^j\}$ such that*

$$L'_{K/k,S}(0, \chi) = c_\chi(h \cdot \eta); \quad (8)$$

3. *the gauge-invariant orbit condition holds:*

$$\frac{L'_{K/k,S}(0, \chi)}{c_\chi(\eta)} \in \mu_4. \quad (9)$$

Proof. The implication $1 \Rightarrow 2$ follows from Theorem 3 after inverting the signed element if necessary. Lemma 2 gives $2 \Rightarrow 3$.

Assume (3). The values of χ on the signed group are all of μ_4 , so choose h with

$$\chi(h) = \frac{L'_{K/k,S}(0, \chi)}{c_\chi(\eta)}$$

and set $\bar{\epsilon} = h \cdot \bar{\eta}$. Then (8) holds. We now verify the three propagation steps needed for the full Stark conjecture.

Step 1: even-component vanishing on both sides. Let ν be the trivial or quadratic character. It is even, so $\nu(\tau) = 1$. Since $\bar{\eta} \in \bar{U}_K^-$, pairing g with $g\tau$ in (1) gives

$$c_\nu(\eta) = 0, \quad c_\nu(\epsilon) = 0.$$

For the analytic side, the one-place signature gives

$$L'_{K/k,S}(0, \nu) = 0$$

for both even characters; this is the even-character vanishing used explicitly in the proof of Roblot's Theorem 6.1 [1, p. 405]. Thus the Stark identity holds on both even components.

Step 2: conjugate-pair determination and Fourier inversion. All logarithms in (1) are real, hence

$$c_{\bar{\chi}}(\epsilon) = \overline{c_\chi(\epsilon)}.$$

Artin L -functions satisfy

$$L'_{K/k,S}(0, \bar{\chi}) = \overline{L'_{K/k,S}(0, \chi)}.$$

Therefore (8) also gives the identity for $\bar{\chi}$. We now have the Stark identity for all four characters of C_4 . Fourier inversion on G determines each component of the logarithmic vector

$$\left(\frac{1}{2} \log |\epsilon^g|_w \right)_{g \in G},$$

so the complete logarithmic part of the rank-one Stark conjecture holds.

Step 3: inheritance of the abelian condition. Choose an actual representative

$$\epsilon = \rho \eta^{h^{-1}}, \quad \rho \in \{\pm 1\}. \quad (10)$$

Conjugation and inversion preserve the abelian extension $K(\sqrt{\eta})/k$ from (4): a k -abelian extension is stable under G , and adjoining a square root of η^{-1} gives the same quadratic extension. If $\rho = 1$, this proves $K(\sqrt{\epsilon})/k$ is abelian. If $\rho = -1$, then

$$K(\sqrt{\epsilon}) \subseteq K(\sqrt{\eta})k(i).$$

The extension $k(i)/k$ is quadratic abelian because k is totally real, and a compositum of abelian extensions is abelian. Hence $K(\sqrt{\epsilon})/k$ is abelian in either case. The representative is a global unit because $\eta \in U_K$.

The logarithmic identity and the abelian square-root condition are exactly the abelian rank-one Stark conjecture with $e = 2$, as proved in Lemma 1. Thus $3 \Rightarrow 1$. $\square$

4 Numerical and circulation boundaries

The five archived controls provide only the following OBSERVED statement: after testing both conjugate character orientations, each independently constructed weak solution matches one fourth-root rotation against certified L' -balls. The old oriented labels were selected after inspecting the analytic target and are withdrawn pending exact Artin transport. They are not used in either proof above.

A future rigorous screen can refute (9) if the enclosed analytic value is separated from all four enclosed rotations of the weak coefficient. Ball overlap is only CERTIFIED NUMERICAL CONSISTENCY, never proof of equality. No such census screen is contained in this note.

Circulation status. This is a local research draft. It must not be submitted, posted, or circulated before the preregistered B3 screen exists as its data section. All email, messaging, submission, and other outbound actions are human-only.

AI-assistance disclosure

AI systems assisted with proof auditing, convention checks, code, and editing. The author reviewed the mathematical argument, checked the cited hypotheses, and is responsible for the claims.

References

- [1] X.-F. Roblot, *Index formulae for Stark units and their solutions*, Pacific J. Math. **266** (2013), no. 2, 391–417, doi:10.2140/pjm.2013.266.391.
- [2] J. Tate, *Les conjectures de Stark sur les fonctions L d'Artin en $s = 0$* , Progress in Mathematics 47, Birkhäuser, Boston, 1984.